Coding & Solution

Date	1 July 2026
Team ID	xxxxxx
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1.Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.

2.The solution should address the core problem statement and implement the key functional requirements.

3.Include all source files, configuration files, and setup instructions needed to run the application.

4.Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/abithachokka
Programming Language(s)	Python
Framework(s) Used	Flask, Scikit-learn, Pandas, NumPy, Matplotlib
Key Features Implemented	Data preprocessing, machine learning model training, flood prediction system, result visualization, accuracy evaluation
Pending / Incomplete Features	Real-time API integration and live flood alert notification system
Setup / Run Instructions	Install Python libraries → Load dataset → Run preprocessing → Train model → Execute prediction module → View results

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The project solution is developed using Python-based machine learning techniques to predict flood occurrences from historical environmental datasets. The codebase is structured into separate modules for preprocessing, model training, prediction, and result analysis. The solution successfully demonstrates accurate flood prediction capability and can be extended further with real-time monitoring and early warning alert systems.